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# Zev-7 & Metronome

## Nixie & Numitron Clocks

### Operator's Guide

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**June 2026**

This manual describes the installation, configuration, and day-to-day operation of the Zev-7 and Metronome clocks. It covers the Nixie-tube models (Zev-7, Zev-7 Mini, Metronome) and the numitron-tube models (Zev-7N, Zev-7N Mini). Throughout this manual all models are referred to as Zev-7, and "Nixie" refers to the time display in general; see Section 2 for how the Zev-7N numitron models differ.

**Revision/Update Information:**

V2 — Rules system replaces Period Overrides;  
adds Zev-7N numitron variants

**Clock Firmware:**

V2

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# Table of Contents

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Quick Start..... QS-1

1. Overview..... 1-1

2. Models..... 2-1

3. Display..... 3-1

4. Buttons..... 4-1

5. Power-On Functions..... 5-1

6. AP Configuration Mode..... 6-1

7. Live Settings Web Interface..... 7-1

8. Rules..... 8-1

9. LED Effects..... 9-1

10. Digit Change Effects..... 10-1

11. Neon (Colon) Display Modes..... 11-1

12. Info Display Cycle..... 12-1

13. Sensors..... 13-1

14. Cathode Anti-Poisoning..... 14-1

15. OTA Firmware Update..... 15-1

16. Factory Reset..... 16-1

17. Troubleshooting..... 17-1

# Quick Start

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Follow these steps to set up the clock for the first time. A full explanation of every option is in the numbered sections that follow.

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## 1. Power on

Connect the clock to its 12V DC power adapter. On a brand-new clock, the display shows a single lit tube bouncing left and right across NX1 through NX7 (Zev-7) or NX1 through NX15 (Metronome), with the digit counting 0–9 as it bounces — this is AP configuration mode. On a previously-used clock that already has WiFi credentials, skip to step 5 — to reconfigure WiFi, hold SW2 for 5 seconds to enter AP mode. To factory reset all settings, hold SW1 and SW2 and apply power. Hold buttons until the display starts.

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## 2. Display configuration

**NOTICE:** If the display shows 00, the processor must be returned to the manufacturer for reprogramming.

Several clock variants are available. The processor is shipped programmed for each specific configuration.

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## 3. Connect to the clock's WiFi

With the AP-mode bouncing tube visible, from a phone or computer, open the WiFi settings and connect to the network named Zev-7 (or Zev-7M for the Mini, ZEV-7N / ZEV-7NM for the numitron models, or Metronome for the Metronome model). No password is required.

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## 4. Configure in a browser

Open a web browser and go to 192.168.4.1. The setup page loads. At minimum:

- On the Network page, select the home WiFi network from the SSID list and enter the password.
- On the Time Zone page, confirm the UTC offset for the local region (US Eastern is 5 hours Minus by default).

Tap Save, then tap Back, then tap Start. The clock saves the settings and reboots.

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## 5. Wait for time to appear

During boot, NX1 briefly shows a scrolling digit "1" while WiFi is connecting, then a scrolling digit "2" while the clock fetches time from the internet. The time appears within a few seconds of connecting to WiFi.

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## 6. Daily use

| To do this                                            | Do this                                                                            |
|-------------------------------------------------------|------------------------------------------------------------------------------------|
| Change brightness temporarily                         | Click SW2 to cycle through brightness levels 1–9                                   |
| Wake the clock when a rule has turned the display off | Click SW1                                                                          |
| See the clock's IP address                            | Double-click SW1 — the address scrolls across the tubes                            |
| Adjust any setting live                               | Browse to <code>http://&lt;ip-address&gt;/</code> from any device on the same WiFi |
| Install new firmware over WiFi                        | Hold SW1 for 10 seconds                                                            |

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## 7. Manual cathode conditioning (Nixie models only)

To recondition the Nixie tubes after shipping or storage, run a one-time 5-minute conditioning cycle: remove power, hold SW2, then reapply power while keeping SW2 held. Release once the digits start cycling. The clock will rapidly cycle every digit on every tube at full brightness for 5 minutes, then continue with normal startup. See Section 14 for full details.

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## 8. Recommended next steps

- Visit the live web interface to explore LED effects, neon patterns, and digit change animations.
- Configure Rules on the live page to dim or turn off the display during sleeping hours, schedule LED lighting changes throughout the day, and disable the motion sensor at chosen times.
- If the optional weather sensor is installed, enable Temperature, Pressure, and Humidity on the Sensors page.

# 1. Overview

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The Zev-7N is a numitron-tube variant of the Zev-7. It is functionally identical to the Zev-7 and runs the same firmware, but its seven displays are DR2010 numitron filament tubes driven by TLC59116 constant-current drivers rather than high-voltage Nixie tubes. Because numitron filaments do not suffer cathode poisoning, the Zev-7N has no anti-poison feature and no high-voltage circuitry. See Section 2 for the full list of differences.

Zev-7 is a seven-tube Nixie clock and the Metronome is its fifteen-tube companion. Both are driven by an ESP32 microcontroller and share the same firmware, web interface, and rule-based scheduling. They display hours, minutes, seconds, and tenths of a second using IN-14 Nixie tubes with HV5530 shift-register drivers. The clocks feature RGB backlighting behind the tubes, neon colon indicators, WiFi connectivity with NTP time sync, an optional motion sensor, an optional light sensor, and (Zev-7 only) an optional weather sensor and under-light strip.

The Metronome is similar to the original functioning of the Metronome clock at Union Square in New York City. It add several additional functions including colored LED lighting, and optional date and weather display.

All settings are stored in non-volatile memory (NVS) and survive power cycles. Configuration is performed through a WiFi access point setup page or a live web interface accessible from any device on the same network.

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## 2. Models

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Several model variants share this firmware. The model is set at the factory and cannot be changed by the user. The Zev-7 and Zev-7 Mini are Nixie-tube clocks; the Zev-7N and Zev-7N Mini are the equivalent numitron-tube clocks. The Metronome uses a separate but operationally similar version of the software.

| Model                  | Identifier in AP / page header | Under-light                                                          |
|------------------------|--------------------------------|----------------------------------------------------------------------|
| Zev-7                  | Zev-7                          | 4 RGB LEDs (color and brightness configurable)                       |
| Zev-7 Mini             | Zev-7M                         | None                                                                 |
| Metronome              | Metronome                      | None                                                                 |
| Zev-7N (numitron)      | Zev-7N                         | 4–20 RGB LEDs (color, brightness, count, and chip type configurable) |
| Zev-7N Mini (numitron) | Zev-7NM                        | None                                                                 |

**NOTICE:** If the display shows 00, the processor must be returned to the manufacturer for reprogramming.

## Numitron Variants (Zev-7N and Zev-7N Mini)

The Zev-7N runs the same firmware as the Zev-7 and shares every feature described in this manual — Rules, LED effects, colon display patterns, the info display cycle, sensors, OTA updates, and the web interface. Wherever this manual says "Nixie" (for example, "Nixie Brightness"), read it as the numitron display on these models. The differences are:

- Display technology: seven DR2010 numitron seven-segment filament tubes driven by four TLC59116 constant-current LED drivers over I2C, in place of the Zev-7's IN-14 Nixie tubes and HV5530 high-voltage shift drivers. The "Nixie Brightness" setting controls numitron filament brightness on these models.
- No high voltage: the numitron display runs at low voltage with no mercury and no ~170 V supply. The high-voltage safety note in the Copyright and Disclaimer section applies to the Nixie models only.
- No cathode anti-poisoning: filaments do not poison, so Section 14 (Cathode Anti-Poisoning), the Anti-Poison settings on the AP and live forms, the power-on conditioning runs, and the noap rule keyword have no effect on Zev-7N models. The display simply runs the time continuously.
- Colon indicators: the colon separators are LEDs rather than neon lamps, but every colon (neon) display pattern in Section 11 works identically.
- Models and addressing: Zev-7N (full, with 4–20 configurable under-light LEDs) and Zev-7N Mini (no under-light). In AP mode they appear as the WiFi networks ZEV-7N and ZEV-7NM, and are reachable by name at zev-7n.local and zev-7nm.local respectively.

## 3. Display

The Zev-7 has seven Nixie tubes; the Metronome has fifteen. Both have a set of neon indicator lamps:

| Position | Tube | Shows              |
|----------|------|--------------------|
| Left     | NX1  | Hours tens         |
|          | NX2  | Hours ones         |
|          | NX3  | Minutes tens       |
|          | NX4  | Minutes ones       |
|          | NX5  | Seconds tens       |
|          | NX6  | Seconds ones       |
| Right    | NX7  | Tenths of a second |

Neon indicators between the tube pairs serve as colon separators (H:M and M:S) and status indicators. A separate "BR" neon is used for additional indication and blinks off briefly if the clock loses its WiFi connection.

The tenths-of-a-second digit updates live at 10 Hz.



The Metronome display adds an additional eight displays. The following tubes count down the time to midnight.

| Position | Tube | Shows              |
|----------|------|--------------------|
| Center   | NX8  | Tenths of a second |
|          | NX9  | Second ones        |
|          | NX10 | Second tens        |
|          | NX11 | Minutes ones       |
|          | NX12 | Minute tens        |
|          | NX13 | Hour ones          |
|          | NX14 | Hour tens          |
| Right    | NX15 |                    |

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## 4. Buttons

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The clock has two buttons: SW1 (left) and SW2 (right).

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### SW1 — During Normal Operation

| Action          | Function                                                                         |
|-----------------|----------------------------------------------------------------------------------|
| Single click    | Wake the display when an active rule has dimmed or turned it off (Rules Timeout) |
| Double-click    | Scroll IP address across the tubes                                               |
| Hold 10 seconds | Start OTA firmware update                                                        |

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### SW2 — During Normal Operation

| Action         | Function                                                    |
|----------------|-------------------------------------------------------------|
| Single click   | Cycle Nixie brightness 1–9 (temporary, not saved to memory) |
| Triple-click   | Enter light sensor calibration mode (~30 seconds)           |
| Hold 5 seconds | Restart in AP configuration mode                            |

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### Button Override

Pressing SW1 temporarily cancels any active rule (for example, a rule that has dimmed or turned off the display overnight) and restores the standard display brightness, LED, and under-light settings. The wake lasts for the configured Rules Timeout (15 Seconds, 1 Minute, or 5 Minutes). When the timeout expires, the active rule re-applies.

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## 5. Power-On Functions

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On Zev-7N numitron models the SW1-only and SW2-only power-on functions below do nothing (there are no cathodes to condition); only the SW1 + SW2 factory reset applies.

Special functions are available by holding buttons while applying power. These are read before the clock initializes its normal button handling.

| Buttons Held | Function                                                                                                  |
|--------------|-----------------------------------------------------------------------------------------------------------|
| SW1 only     | Startup display test and cathode exercise (5 min). Cycles all digits 0–9 on all tubes at full brightness. |
| SW2 only     | Anti-poison conditioning run (5 min). Rapidly cycles all cathodes to recondition tubes.                   |
| SW1 + SW2    | Factory reset. Clears all saved settings and restarts into AP configuration mode.                         |

## 6. AP Configuration Mode

### Entering AP Mode

- Hold SW2 for 5 seconds during normal operation, OR
- Apply power with SW2 held down

AP mode is indicated by a single lit tube that bounces left and right across NX1 through NX7 (Zev-7) or NX1 through NX15 (Metronome), with the digit counting 0–9 as it bounces.

### Connecting

1. Connect a phone or computer to the WiFi network named Zev-7 (or Zev-7M, ZEV-7N, ZEV-7NM, or the custom AP name).
2. Open a web browser and go to 192.168.4.1
3. Configure settings and press Save. Press Back, and press Start.

The setup form is split across multiple pages (Network, Display, Time Zone, Sensors, Rule Editor, etc.). Use the page buttons on the home page to navigate.

### Network Settings

| Setting              | Description                                                                            |
|----------------------|----------------------------------------------------------------------------------------|
| Access Point Name    | WiFi network name when in AP mode (default: Zev-7 or Zev-7M depending on model)        |
| SSID                 | Select the home WiFi network from the scan list                                        |
| SSID Override        | Manually type an SSID if it doesn't appear in the scan                                 |
| Password             | WiFi password                                                                          |
| NTP Server           | Primary time server (default: time.cloudflare.com)                                     |
| Alternate NTP Server | Backup time server, used if the primary becomes unreachable (default: time.google.com) |
| Live Form Password   | Optional password for the live web interface. Leave blank for no password.             |
| Live Form Enabled    | Turns the live web interface on or off (default: On)                                   |

### Display Settings

| Setting | Options | Default |
|---------|---------|---------|
|---------|---------|---------|

|                     |                                                          |              |
|---------------------|----------------------------------------------------------|--------------|
| Time Format         | 12 Hour, 24 Hour                                         | 12 Hour      |
| Date Display Format | Off, MM/DD/YY, DD/MM/YY, YY/MM/DD                        | MM/DD/YY     |
| Info Transition     | Fade, Scroll, Wave, Spin Wave                            | Spin Wave    |
| Neon Display        | See Section 11: Neon Display Modes                       | Railroad     |
| Nixie Brightness    | 1–9                                                      | 8            |
| LED Effect          | See Section 9: LED Effects                               | Random Multi |
| RGB LED Brightness  | 0–9                                                      | 4            |
| LED Follows Digit   | On, Off (LED under each tube reflects that tube's digit) | Off          |

## Digit Change Effect

| Option           | Description                                                     |
|------------------|-----------------------------------------------------------------|
| Off              | Instant digit change, no animation                              |
| Crossfade        | Smooth PWM fade between old and new digits                      |
| Slot Machine 10s | Cascading countdown effect every 10 seconds                     |
| Slot Machine 60s | Cascading countdown at the minute boundary with ease-out timing |

Crossfade Duration is independently configurable from 0 (instant) to 9 (900 ms) in 100 ms steps. Default is 5 (500 ms). Slot Machine Speed is configurable 0–9 (default 5); higher values run the cascade faster.

## Anti-Poison Settings (Nixie models only)

| Setting           | Options                                                                                                                                             | Default                |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| Anti-Poison Cycle | Off, 5 Seconds Every Minute, 10 Seconds Every Minute, 1 Minute Every 10 Minutes, 1 Minute Every Hour, 10 Minutes Every Hour, 30 Minutes at Midnight | 5 Seconds Every Minute |
| Start Second      | 0–59                                                                                                                                                | 30                     |

## Rules Timeout

How long the SW1 button-wake lasts when a rule has dimmed or turned the display off: 15 Seconds, 1 Minute, or 5 Minutes (default: 15 Seconds). When the timeout expires, the active rule re-applies.

## Time Zone

| Setting              | Options     | Default |
|----------------------|-------------|---------|
| UTC Offset Hours     | 0–23        | 5       |
| UTC Offset Minutes   | 0–59        | 0       |
| UTC Offset Direction | Minus, Plus | Minus   |

## Daylight Saving Time

| Setting              | Options     | Default |
|----------------------|-------------|---------|
| DST Start Month      | Jan–Dec     | March   |
| DST Start Week       | 1–5         | 2       |
| DST Start Day        | Sun–Sat     | Sunday  |
| DST Start Hour       | 0–23        | 2       |
| DST Offset Hours     | 0–13        | 1       |
| DST Offset Direction | Plus, Minus | Plus    |

## Standard Time

| Setting         | Options | Default  |
|-----------------|---------|----------|
| STD Start Month | Jan–Dec | November |
| STD Start Week  | 1–5     | 1        |
| STD Start Day   | Sun–Sat | Sunday   |
| STD Start Hour  | 0–23    | 2        |

Default DST/STD settings are configured for US Eastern Time.

## Weather Sensor (BME280, Zev-7 only)

| Setting             | Options                          | Default    |
|---------------------|----------------------------------|------------|
| Temperature Display | Off, Fahrenheit, Celsius, Kelvin | Fahrenheit |
| Temperature Offset  | –15 to +15 (1° increments)       | 0          |
| Display Pressure    | Off, Inches Mercury, Millibars   | Off        |
| Display Humidity    | Off, Yes                         | Off        |

## Light Sensor

| Setting | Options | Default |
|---------|---------|---------|
|---------|---------|---------|

|                          |           |     |
|--------------------------|-----------|-----|
| Light Level Threshold    | Off, 1–50 | Off |
| Minimum Nixie Brightness | 0–9       | 4   |
| Minimum LED Brightness   | 0–9       | 4   |

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### Motion Sensor (PIR)

| Setting         | Options                                                        | Default |
|-----------------|----------------------------------------------------------------|---------|
| Running Timeout | Off, 10s, 20s, 30s, 1 min, 5 min, 10 min, 15 min, 30 min, 1 hr | Off     |
| Rules Timeout   | Off, 10s, 20s, 30s, 1 min, 5 min, 10 min, 15 min, 30 min, 1 hr | Off     |

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### Under Light (Zev-7 only)

| Setting    | Options          | Default       |
|------------|------------------|---------------|
| Brightness | 0–9              | 4             |
| Color      | RGB color picker | Blue (0000FF) |

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### Miscellaneous

| Setting                      | Options | Default                                                                               |
|------------------------------|---------|---------------------------------------------------------------------------------------|
| Date Display Start Second    | 0–59    | 20                                                                                    |
| Weather Display Start Second | 0–59    | 40                                                                                    |
| Firmware Update URL          | URL     | <a href="https://mitchsf.github.io/Firmware/">https://mitchsf.github.io/Firmware/</a> |

The Date and Weather start seconds run independently. Keep them at several seconds apart so a long info cycle on one doesn't collide with the other.

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## 7. Live Settings Web Interface

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Once connected to WiFi, the live settings page is accessible from any device on the same network.

Browse to `http://<clock-ip-address>/` To find the IP address, double-click SW1 — the address scrolls across the tubes.

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### Using a Name Instead of an IP Address (mDNS)

On many networks the clock can be reached by name instead of by IP address. Browse to `http://zev-7.local/` (or `http://zev-7m.local/` for the Mini, `http://zev-7n.local/` or `http://zev-7nm.local/` for the numitron models, or `http://metronome.local/` for the Metronome) and the page should load. The hostname is derived from the Access Point Name set during configuration, lowercased, with any spaces or special characters converted to hyphens. For example, an AP Name of "Zev-7" becomes `zev-7.local`; "My Office" becomes `my-office.local`.

If the Access Point Name has been customized during AP configuration, the mDNS address follows the same rule: the new AP Name (lowercased, with any non-alphanumeric characters converted to hyphens) followed by `.local`. This makes it easy to give each clock in a multi-clock environment a unique address — for instance, changing the AP Name to "Kitchen Clock" on one unit makes that clock reachable at `http://kitchen-clock.local/` regardless of its IP.

This feature relies on mDNS (also known as Bonjour or zero-configuration networking) and does not work on every network. Some routers — particularly older models or routers with multicast filtering enabled — block the discovery messages mDNS depends on, in which case the `.local` address will not resolve. Some browsers and operating systems also handle mDNS inconsistently: macOS, iOS, and Windows 10/11 generally work out of the box, while some Android versions and Linux distributions require additional setup. If the `.local` address does not load, fall back to the IP address (double-click SW1 to display it).

If a Live Form Password was set during AP configuration, the page prompts for it before any settings are shown.

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### Available Controls

All changes take effect immediately without rebooting. Use the Save to Memory button to persist changes across power cycles. The form is split across multiple pages. The address bar shows which page is currently displayed. Some settings are only available on the Settings pages. These are accessed by holding SW2 for five seconds.

- Nixie Brightness (1–9)
- RGB LED Brightness (0–9)
- Neon Display (all 9 patterns)
- LED Effect (all 13 effects); LED Color picker (active when LED Effect is Custom)



- Digit Change Effect, Crossfade Duration, Slot Machine Speed
- Info Transition (Fade, Scroll, Wave, Spin Wave)
- Date Display (Enable/Disable; format is set on the Settings form)
- Temperature Display, Offset, Pressure, Humidity
- Anti-Poison Cycle, Start Second
- Motion Sensor: Running Timeout, Rules Timeout
- Light Sensor settings
- Under Light Brightness and Color (Zev-7 only)
- LED Follows Digit
- Date Display Start Second, Weather Display Start Second
- Rule Editor page (add, edit, and delete time-range rules)
- Disable Rules toggle (temporarily ignores all rules without deleting them)
- Show IP Address button

## 8. Rules

Rules schedule different display, LED, under-light, and motion-sensor settings for specific times of day or days of the week. Up to 32 rules can be defined. Each rule specifies a time range and one or more overrides — for example, "turn the display off from 11:00 PM to 7:00 AM" or "set the LEDs to red at brightness 9 from 6:00 PM to 10:00 PM on weekdays."

**Priority:** when two rules overlap in time, the rule listed higher in the schedule wins for the overlapping window. Use the up and down arrows on the right of each row to change a rule's priority (up = higher priority). The Rule Editor flags overlapping rules with a warning icon (⚠) so unintended interactions are visible at a glance.

The Disable Rules toggle on the Rule Editor page temporarily ignores all rules without deleting them. This is useful for making live adjustments without rules re-applying. The toggle does not persist across reboots and always defaults to Off (rules active) when the live page is opened.

**Rule Editor page.** Tap Rule Editor on the live form home page to manage the rule list. Each rule is written as a short text expression and appears in the schedule table below the input field. Tap a row in the schedule to load that rule into the editor; tap Apply to save it. Tap Delete to remove the loaded rule.

**Syntax.** A rule starts with a time range (`t=...`) followed by any combination of field overrides, separated by commas or spaces. Time ranges use a 24-hour clock and the end time is exclusive (`t=5-7` covers 5:00 through 6:59). Times may be written as a bare hour (`t=5` → 5:00-5:59), an hour range (`t=5-7`), exact minutes (`t=5:30-6:00`), or in compact four-digit form (`t=0530-0600` = 5:30-6:00). Cross-midnight ranges are allowed: `t=22-6` means 10:00 PM through 5:59 AM.

**Field keys.** Each override uses a 2-letter short key (or its full descriptive name) followed by "=" and a value. The supported keys are: `nb` / `nixiebrightness` (Nixie brightness 0-9), `cd` / `colondisplay` (colon neon pattern 0-8, see Section 11), `lb` / `ledbrightness` (LED brightness 0-9), `le` / `ledeffect` (LED effect 0-12, see Section 9), `lc` / `ledcolor` (LED color, hex #RRGGBB or color name), `ub` / `underbrightness` (under-light brightness 0-9, Zev-7 full only), `uc` / `undercolor` (under-light color, Zev-7 full only), and `pir` / `passiveinfrared` (motion-sensor timeout 0-9 - same scale as the Motion Sensor Timeout dropdown: 0=Off, 1=10s, 2=20s, 3=30s, 4=1m, 5=5m, 6=10m, 7=15m, 8=30m, 9=1hr). Add the bare keyword `"wd"` / `"weekday"` to restrict the rule to weekdays (Mon-Fri) or `"we"` / `"weekend"` to restrict it to weekends (Sat-Sun). Fields may appear in any order - the time range does not need to be first.

**Note:** `lc` is honored only when `le=12` (Custom). Using `lc` by itself auto-promotes the rule to effect 12. Pairing `lc` with any other `le` value is rejected by the rule editor.

### Keyword Reference

**Keyword reference.** Each rule may use any combination of the following keys. The short and long forms are interchangeable.

| Keyword | Description |
|---------|-------------|
|---------|-------------|

| t / time              | Time range. Specifies when the rule is active. Required in every rule. Format: HH:MM-HH:MM, H-H, HHMM-HHMM, or bare H. End time is exclusive.                                                                                                                                                  |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| nb / nixiebrightness  | Nixie tube brightness (0-9, 0 = display off).                                                                                                                                                                                                                                                  |
| cd / colondisplay     | Colon neon pattern (0-8). See Section 11 for the pattern catalog.                                                                                                                                                                                                                              |
| lb / ledbrightness    | Digit RGB LED brightness (0-9, 0 = LEDs off).                                                                                                                                                                                                                                                  |
| le / ledeffect        | Digit LED effect (0-12). See Section 9 for the effect catalog. Index 12 (Custom) renders a solid color from the lc value below; any other index runs the named animation.                                                                                                                      |
| lc / ledcolor         | Digit LED solid color. Hex (#RRGGBB) or color name (red, green, blue, white, yellow, cyan, magenta, orange, purple, pink, black). Honored only when le=12 (Custom). Using lc by itself auto-promotes the rule to effect 12; pairing lc with any other le value is rejected by the rule editor. |
| ub / underbrightness  | Under-light brightness (0-9, 0 = off). Zev-7 full only.                                                                                                                                                                                                                                        |
| uc / undercolor       | Under-light color. Same hex or color-name format as lc. Zev-7 full only.                                                                                                                                                                                                                       |
| pir / passiveinfrared | Motion-sensor timeout (0 = Off, 1 = 10s, 2 = 20s, 3 = 30s, 4 = 1m, 5 = 5m, 6 = 10m, 7 = 15m, 8 = 30m, 9 = 1hr).                                                                                                                                                                                |
| wd / weekday          | Bare keyword (no = value). Restricts the rule to weekdays (Monday-Friday).                                                                                                                                                                                                                     |
| we / weekend          | Bare keyword (no = value). Restricts the rule to weekends (Saturday-Sunday).                                                                                                                                                                                                                   |
| noap / noantipoison   | Bare keyword (no = value). Suppresses anti-poison cycles during this rule's window. Useful for quiet/dim windows where a full-brightness anti-poison run would be disruptive. Nixie models only; ignored on Zev-7N numitron models.                                                            |
| off                   | Bare keyword (no = value). Shortcut that expands to nb=0, lb=0, ub=0 (everything off: Nixie tubes, digit LEDs, and under-light). On Zev-7 Mini the ub field isn't registered, so off only expands to nb=0, lb=0.                                                                               |

## Examples

| Rule                    | Time Format                     | What It Does                                                                                                                              |
|-------------------------|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| t=5, nb=9               | H - bare hour                   | At 5:00 AM, set Nixie brightness to maximum. The bare-hour form covers the full 60 minutes (5:00 through 5:59).                           |
| t=22-6, nb=0            | H-H - cross-midnight hour range | Overnight blackout, every day: turns the Nixie display off from 10:00 PM through 5:59 AM. The end time wraps past midnight automatically. |
| t=22:00-06:00, wd, nb=0 | HH:MM-HH:MM - full precision    | Weeknight blackout (Mon-Fri only). Same window as the rule above, written with full HH:MM digits and restricted to weekdays.              |

|                                              |                                       |                                                                                                                                                                                   |
|----------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| t=22-9, we, nb=0                             | H-H - cross-midnight + weekend filter | Weekend sleep-in (Sat-Sun only): display off from 10:00 PM through 8:59 AM the next morning. Pairs naturally with the weeknight rule above.                                       |
| t=5:30-6:00, lb=9, le=1                      | H:M-H:M - minute precision            | Sunrise wake: 30 minutes of Red-White-Blue Wave (effect 1) at LED brightness 9 starting at 5:30 AM.                                                                               |
| t=0530-0600, lb=9, le=1                      | HHMM-HHMM - compact 4-digit           | Identical window to the rule above, written in the no-colon four-digit compact form. Useful when typing on a phone keyboard.                                                      |
| t=17:30-18:30, lb=9, ub=9, uc=orange         | HH:MM-HH:MM - one-hour minute span    | Dinner spotlight, every day: maximum LED and under-light brightness with an orange under-light color, from 5:30 PM through 6:29 PM.                                               |
| t=12-13, lc=green                            | H-H - single-hour range               | Lunch-hour green: tints all digit LEDs solid green from 12:00 PM through 12:59 PM. lc renders a solid color (no effect needed).                                                   |
| t=6-22, pir=0                                | H-H - daytime range                   | Disables the motion sensor during the day (6:00 AM through 9:59 PM) so the display stays on regardless of motion.                                                                 |
| t=7-9, wd, lb=9, le=1                        | H-H + weekday filter                  | Weekday morning energizer (Mon-Fri only): Red-White-Blue Wave at maximum LED brightness from 7:00 AM through 8:59 AM.                                                             |
| t=20-23, we, le=5, lb=9                      | H-H + weekend filter                  | Weekend evening (Sat-Sun only): Random Big Color Wave (effect 5) at full LED brightness from 8:00 PM through 10:59 PM.                                                            |
| t=23:00-23:45, nb=1                          | HH:MM-HH:MM - partial hour            | Final-stretch dim: minimum non-zero Nixie brightness for the last 45 minutes before midnight.                                                                                     |
| time=22:00-06:00, weekday, nixiebrightness=0 | Full keyword names                    | Identical to the weeknight blackout shown earlier, but written with the full descriptive keyword names instead of the short keys. The parser accepts either form interchangeably. |
| nixiebrightness=2, weekend, time=22:00-09:00 | Order shuffled - time is not          | Weekend overnight dim: Nixie brightness drops to 2 from 10:00                                                                                                                     |

|  |       |                                                                                                                            |
|--|-------|----------------------------------------------------------------------------------------------------------------------------|
|  | first | PM through 8:59 AM, Saturday and Sunday only. Fields may appear in any order - the time range does not have to come first. |
|--|-------|----------------------------------------------------------------------------------------------------------------------------|

## 9. LED Effects

The Zev-7 has 7 RGB LEDs behind the Nixie tubes (one per tube); the Metronome has 15 RGB LEDs (one per tube). The Zev-7 also has separate under-light LEDs along the front edge of the base — 4 by default, configurable from 4 to 20 in the AP Configuration form, with a choice of APA102 or WS2812B chip types. The Metronome has no under-light LEDs. The under-light LEDs have independent brightness and color control and are not affected by LED effects.

LED effects apply to the 7 backlight LEDs:

| #  | Effect                | Description                                                                                          |
|----|-----------------------|------------------------------------------------------------------------------------------------------|
| 0  | Off                   | All LEDs dark                                                                                        |
| 1  | Red-White-Blue Wave   | Red, white, and blue cycling with sine-wave modulation                                               |
| 2  | Blue-White Wave       | Blue and white alternating wave                                                                      |
| 3  | Red-White Wave        | Red and white alternating wave                                                                       |
| 4  | Random All Color Wave | Single random color with sine-wave modulation                                                        |
| 5  | Random Big Color Wave | Random colors per LED with sine-wave modulation (default)                                            |
| 6  | Random Bright Color   | Bright saturated random colors with sine-wave modulation                                             |
| 7  | Red Wave              | Red with sine-wave fade                                                                              |
| 8  | Green Wave            | Green with sine-wave fade                                                                            |
| 9  | Blue Wave             | Blue with sine-wave fade                                                                             |
| 10 | Larson Scanner        | Knight Rider effect cycling through 7 colors                                                         |
| 11 | Cycle All             | Rotates through effects 1–10, changing every 10 seconds                                              |
| 12 | Custom                | Renders the color set in the LED Color picker (or by the lc rule keyword) as a solid across all LEDs |

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## 10. Digit Change Effects

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When a time digit changes, the clock can animate the transition:

| Effect           | Description                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------|
| Off              | Instant change, no animation                                                                                  |
| Crossfade        | Smooth PWM fade from old digit to new digit. Duration configurable 0–900 ms in 100 ms steps (default 700 ms). |
| Slot Machine 10s | Cascading countdown effect triggered every 10 seconds                                                         |
| Slot Machine 60s | Cascading countdown at the minute boundary with ease-out timing                                               |

Slot Machine Speed (0–9, default 5) tunes how fast the cascade runs. 0 makes the change instant; higher values run the cascade faster. Crossfade speed can also be changed in the same way.

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## 11. Neon (Colon) Display Modes

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The neon indicator lamps between the tubes can display various patterns:

| # | Mode          | Description                                     |
|---|---------------|-------------------------------------------------|
| 0 | Off           | All neons dark                                  |
| 1 | All On        | All neons continuously lit                      |
| 2 | Bottom Flash  | Bottom colon neons blink each second            |
| 3 | All Flash     | All neons blink each second                     |
| 4 | Railroad      | Alternating railroad-crossing pattern (default) |
| 5 | Strobe        | Horizontal stripe alternating pattern           |
| 6 | Left to Right | Sweep pattern across neon positions             |
| 7 | AM/PM Top Off | AM/PM indicator with top neons off              |
| 8 | AM/PM Top On  | AM/PM indicator with top neons on               |

## 12. Info Display Cycle

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The clock periodically interrupts the time display to show additional information. Date and weather information run on independent schedules controlled by the Date Display Start Second and Weather Display Start Second settings (each fires once per minute at the configured second).

---

### Information Shown (if enabled)

Date: Displayed as MM/DD/YY, DD/MM/YY, or YY/MM/DD depending on configuration.

Temperature: Shown in tenths of a degree (e.g., 78.2). Temperature units selectable as Fahrenheit, Celsius, or Kelvin. Negative temperatures are indicated by a special neon pattern.

Barometric Pressure: Displayed in inches of mercury or millibars.

Humidity: Displayed as a percentage with tenths (e.g., 55.2%).

Each item is held on the display briefly, then transitions to the next using the configured Info Transition (Fade, Scroll, Wave, or Spin Wave). The first info trigger after boot is suppressed so the clock has a chance to settle.



## 13. Sensors

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### Temperature / Pressure / Humidity (BME280, Zev-7 only)

If a BME280 sensor is installed, the clock can display temperature, barometric pressure, and humidity as part of the info display cycle. A calibration offset is available for temperature.

The sensor is auto-detected at startup on the I2C bus (addresses 0x76 and 0x77).

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### Light Sensor

Calibration: Triple-click SW2 to enter calibration mode. The raw sensor reading (0–99) is displayed on the leftmost two tubes for ~30 seconds.

Calibration: Triple-click SW2 to enter calibration mode. The raw sensor reading (0–99) is displayed on the leftmost two tubes for ~30 seconds.

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### Motion Sensor (PIR)

When enabled, the display goes dark after the configured timeout with no detected motion. When motion is detected, the display smoothly fades in to the configured brightness. When motion stops and the timeout expires, the display smoothly fades out.

Two separate timeout settings are available: Running Timeout controls PIR behavior during normal operation, and Rules Timeout controls PIR behavior while a rule has turned the display off. Each can be configured independently. If Rules Timeout is Off (default), the display remains in its rule-driven state regardless of motion. Individual rules can also set their own per-rule PIR timeout with the `pir=` field (see Section 8).

## 14. Cathode Anti-Poisoning

Applies to Nixie models only. The Zev-7N numitron variants use filament seven-segment tubes that do not suffer cathode poisoning. On those models this entire section, the Anti-Poison settings, the power-on conditioning runs, and the noap rule keyword have no effect.

Nixie tubes can develop "cathode poisoning" when certain digits are displayed for long periods without exercising other cathodes. The unused cathodes accumulate a chemical deposit that gradually reduces their brightness and produces a faded or "ghosted" appearance when those digits are eventually shown. The anti-poison feature periodically cycles all 10 digits on every tube at high speed to prevent and reverse this effect.

**Anti-Poison Cycle** is a single setting that picks a paired duration and schedule. The available cycles are 5 Seconds Every Minute, 10 Seconds Every Minute, 1 Minute Every 10 Minutes, 1 Minute Every Hour, 10 Minutes Every Hour, and 30 Minutes at Midnight.

**Trigger conditions:** all cycles fire at the configured Start Second. The five non-Midnight cycles only run when the tubes are currently on — if an active rule has the display off, or the PIR motion timeout has blanked it, the cycle is skipped that interval. The Midnight cycle is the exception: it always runs, even when the display is otherwise off.

**During a cycle:** tubes run at full brightness regardless of any active rule. RGB LEDs continue running their configured effect and brightness. With LED Follows Digit enabled, the under-tube LEDs follow whichever tubes are lit — since the cycle keeps all tubes lit, the LEDs stay on through the cycle. Both colon neons are lit. If PIR turns the tubes off mid-run on a non-Midnight cycle, the cycle aborts cleanly and the display goes back to its off state.

Smart conditioning algorithm: rather than spending an equal amount of time on every digit, the firmware tracks which cathodes get exercised by the normal time display and weights the anti-poison cycle to favor the ones that do not. For example, in 12-hour mode the leftmost hour tube only ever displays a "1", so digits 0 and 2 through 9 on that tube get roughly four times the conditioning time of digit 1. The minute and second tens tubes never display 6 through 9 during normal use, so those digits also get extra time. The algorithm uses separate weighting tables for 12-hour and 24-hour modes, so changing the time format automatically rebalances the cycle.

The Anti-Poison Start Second setting controls when in the minute the cycle begins. This can be offset from the Date and Weather Display start seconds to avoid conflicts. When running multiple clocks, different start seconds can synchronize or stagger the cycles.

### Manual Conditioning (Hold SW2 at Power-On)

A separate, more aggressive 5-minute conditioning cycle is available as a power-on function. This is intended for tubes that are visibly faded, have been in storage for a long time, or have not been exercised by the regular anti-poison schedule.

To start a manual conditioning run:

1. Remove power from the clock.
2. Press and hold SW2 (right button).
3. Apply power while continuing to hold SW2.
4. Release SW2 once the conditioning cycle begins.

All seven tubes will rapidly cycle through every digit 0–9 at full brightness for 5 minutes. Both colon neons remain lit throughout the run. The clock does not connect to WiFi or fetch the time during this period — it focuses entirely on conditioning the cathodes.

After 5 minutes the cycle ends automatically: the display clears, the neons turn off, and the clock proceeds with its normal boot sequence (WiFi association, NTP sync, then time display). Power can be removed at any point to stop the cycle early; no settings are changed by this operation.

Recommended use: run manual conditioning when first installing the clock, after the clock has been unplugged for several weeks, or any time the tubes show visibly faded digits that the scheduled anti-poison has not corrected. There is no harm in running it more than once.

## 15. OTA Firmware Update

The clock can update its firmware over WiFi. To trigger an update, hold SW1 for 10 seconds.

The firmware can also be updated from the web settings page, without using the button. On the System page, press Update Firmware. The settings form is replaced by a status message while the clock checks the firmware server. When the check completes, the result is shown — either confirmation that the firmware is already current (with its version number) or a brief error — and the page then returns to the home screen automatically. If a newer version is available, it is downloaded and installed and the clock reboots into it. The tubes show the same update status described below regardless of whether the update is started from the button or the web page.

### Display Feedback

All OTA status appears on the leftmost two tubes (NX1-NX2). All other tubes are blank during the update.

| Display                      | Meaning                                                                       |
|------------------------------|-------------------------------------------------------------------------------|
| 88                           | Connecting to WiFi and the firmware server                                    |
| Version number (e.g., 1 0 7) | Firmware is already current. Displayed for 3 seconds, then returns to normal. |
| NX1 counts 0–9               | Downloading and flashing new firmware (cycling counter)                       |
| 10                           | Update successful. Reboots automatically after 3 seconds.                     |

### Error Codes

If the update fails, NX1-NX2 display a two-digit error code, then the clock reboots after 3 seconds:

| Code | Meaning                                                       |
|------|---------------------------------------------------------------|
| 91   | No WiFi connection or firmware URL not configured             |
| 92   | Could not fetch the VERSION file from the server              |
| 93   | VERSION file is malformed (missing or invalid version string) |
| 94   | Binary download or flash write failed                         |

## Version Behavior

Major version change (e.g., 1.x.x to 2.x.x): All settings are reset to factory defaults after the update.

Minor or patch change (e.g., 1.0.6 to 1.0.7): All settings are preserved.

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## 16. Factory Reset

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To reset all settings to factory defaults:

1. Remove power from the clock.
2. Press and hold both SW1 and SW2.
3. Apply power while holding both buttons.
4. Continue holding until the display clears (about 5 seconds).
5. Release both buttons. The clock restarts in AP configuration mode.

All WiFi credentials, display preferences, rules, and sensor settings are erased. The model code (which identifies the unit as a Zev-7 or Zev-7 Mini) is preserved.

## 17. Troubleshooting

| Symptom                                  | Possible Cause                                                         | Solution                                                                                                                                                                   |
|------------------------------------------|------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Display is blank after power on          | Waiting for WiFi association or NTP time sync                          | NX1 scrolls digit "1" while WiFi is connecting and digit "2" while NTP is syncing. If neither completes within 2 minutes, the clock automatically reboots and tries again. |
| Display shows steady 99                  | Clock is in AP config mode                                             | Connect to the Zev-7 or Zev-7M WiFi and configure at 192.168.4.1.                                                                                                          |
| Display shows steady 00 and never starts | Missing or invalid model code                                          | Contact the manufacturer. The unit needs to be re-provisioned.                                                                                                             |
| Tubes are dim                            | An active rule, or the light sensor adjusting for low light            | Press SW1 to temporarily wake the display. Check the active rules on the Rule Editor page and the Light Sensor settings.                                                   |
| BR neon blinks off briefly               | WiFi connection lost                                                   | The clock is attempting to reconnect automatically. If it persists, check the router or re-enter AP mode.                                                                  |
| No WiFi connection at boot               | Wrong credentials or out of range                                      | Enter AP mode (hold SW2 for 5 seconds) and reconfigure WiFi.                                                                                                               |
| Time is wrong                            | Incorrect time zone or DST                                             | Check UTC offset and DST/STD configuration in AP setup.                                                                                                                    |
| Time is briefly wrong, then corrects     | Primary NTP server slow; alternate took over                           | Normal behavior. The clock falls back to the alternate NTP server if the primary is stale for more than 30 minutes.                                                        |
| Digits appear faded or ghosted           | Cathode poisoning                                                      | Enable anti-poison schedule. Hold SW2 at power-on for a 5-minute conditioning run.                                                                                         |
| OTA shows error 91                       | No WiFi or missing URL                                                 | Ensure WiFi is connected and firmware URL is configured.                                                                                                                   |
| LEDs not lighting                        | Brightness set to 0, or an active rule has dimmed or disabled the LEDs | Check RGB LED Brightness on the live page. Rules can override LED brightness and effect — review the Rule Editor.                                                          |
| Live web interface won't load            | Live form disabled or wrong password                                   | Verify Live Form Enabled is On in AP setup. Verify the Live                                                                                                                |

|                                 |                                        |                                                                                                                                           |
|---------------------------------|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
|                                 |                                        | Form Password if one was set.                                                                                                             |
| Under-light not working (Zev-7) | Brightness set to 0, or an active rule | Check Under Light Brightness on the live page. Rules can independently set the under-light brightness and color — review the Rule Editor. |



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# Copyright and Disclaimer

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The Nixie tubes used in this clock contain a small amount of mercury and neon gas under low pressure. Operate the clock at the supplied voltage only; the high-voltage circuitry inside generates approximately 170 volts and should not be opened or serviced by the end user. This applies to the Nixie models only: the Zev-7N numitron variants use low-voltage filament tubes that contain no mercury and generate no high voltage.

Zev-7 Nixie Clock — Designed by Mitchell Feig d/b/a Angry Electrons